FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
CONCEPT RECAPITULATION TEST – III
PAPER –2
TEST DATE: 01-05-2024

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 51 questions.
- Each subject (PCM) has 17 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section-B & Section-C**.
  Section – A *(01 – 04, 18 – 21, 35 – 38)*: This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
  Section – A *(05 –07, 22 – 24, 39 – 41)*: This section contains **NINE (09)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
  Section – B *(08 – 13, 25 – 30, 42 – 47):* This section contains **EIGHTEEN (18)** numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.
  Section – C *(14 –17, 31 – 34, 48 – 51)*: This section contains **SIX (06) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	0	In all other cases.

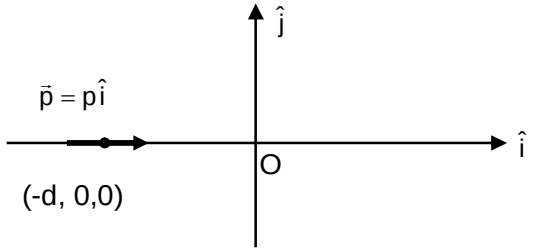
Physics

PART – I

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

1. An electric dipole having dipole moment $\vec{p}$ is placed on the x-axis as shown at (-d, 0, 0). An electron (mass = m, change = e) is having uniform circular motion in the influence of the dipole in y – z plane with x = 0. Then the speed of electron will be. (Neglect gravity)



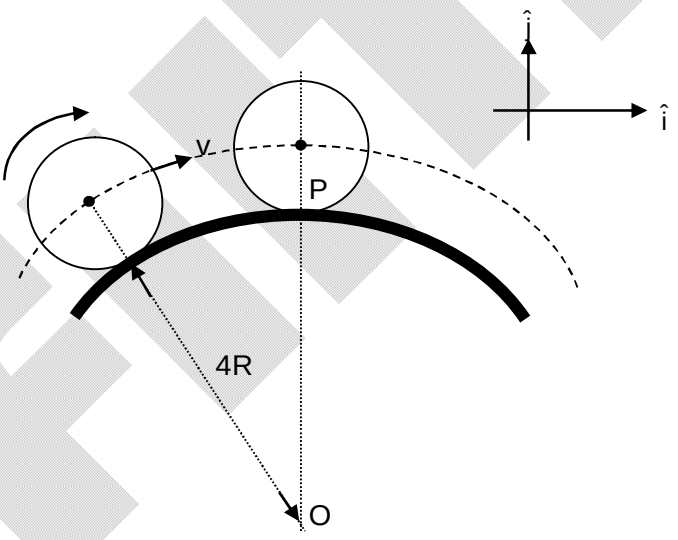
(A) $v = \sqrt{\dfrac{eP}{6\pi \in_0 md^2}}$

(B) $v = \sqrt{\dfrac{\sqrt{2}eP}{\pi \in_0 md^2}}$

(C) $v = \sqrt{\dfrac{eP}{6\sqrt{3}\pi \in_0 md^2}}$

(D) $v = \sqrt{\dfrac{eP}{\sqrt{2}\pi \in md^2}}$

2. A cylinder of radius R is rolling on a circular road (vertical section) of radius 4R as shown with constant speed of v. P is the point of contact of the cylinder with the surface when it is at the highest point of its path. At the given instant shown, acceleration of point P is



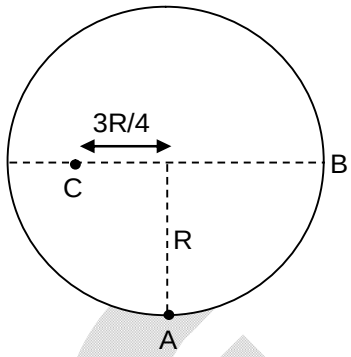
(A) zero

(B) $\dfrac{12v^2}{25R}(\hat{j})$

(C) $\dfrac{4v^2}{5R}\left(-\hat{j}\right)$

(D) $\dfrac{4v^2}{5R}\left(\hat{j}\right)$

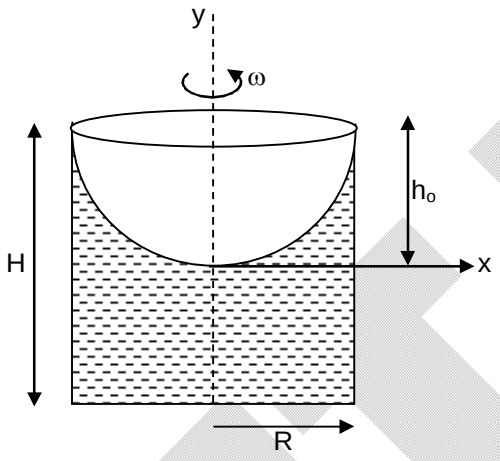
3. A smooth ring of radius R is fixed in horizontal plane. A bead of mass m is threaded into it. Apart from normal force due to ring, it experiences a repulsive force due to a central conservative force whose centre is C. The potential energy of the bead is $U = \frac{K}{x}$, where K is a positive constant and x is its distance from C. The bead is released from rest from A. The normal force when it teaches position B as shown is (Neglect gravity)



(A) $\frac{192K}{245R^2}$ (B) $\frac{32K}{245R^2}$

(C) $\frac{16K}{35R^2}$ (D) $\frac{16K}{49R^2}$

4. An ideal liquid of density ρ is filled in a cylindrical container upto $3/4^{th}$ height. Now it is rotated about a vertical axis passing through its axis of symmetry with constant angular speed ω, such that liquid is on the verge of falling out of the container as shown. Then



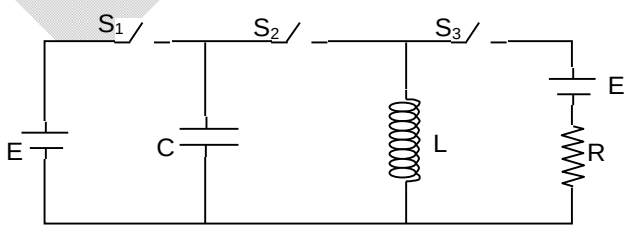
(A) $\omega = \frac{\sqrt{gH}}{2R}$ (B) $\omega = \frac{2\sqrt{gH}}{R}$

(C) $\omega = \frac{\sqrt{2gH}}{H}$ (D) $\omega = \frac{\sqrt{gH}}{R}$

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

5. In the arrangement shown, S_1 and S_3 are closed for a long time. Now at t = 0, S_2 is closed and at the same time S_1 and S_3 are opened. $[E = 2V, L = 2H, C = 2F, R = 2\Omega]$



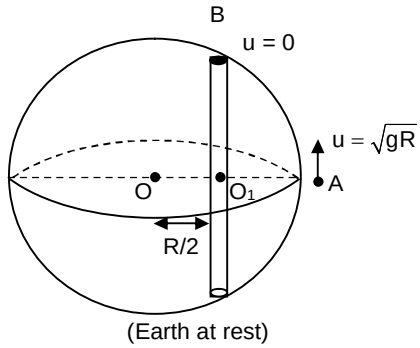
(A) After t = 0, maximum charge on capacitor is 4 coulomb

(B) After t = 0, maximum charge on capacitor is $2\sqrt{5}$ coulomb

(C) After t = 0, maximum current in inductor is $\sqrt{5}$ A

(D) After t = 0, maximum current in inductor is 1A

6. Particle A is projected with a speed $u = \sqrt{gR}$ tangentially very close to surface of earth. At the same time another particle B of same mass is released on surface of earth along the small smooth tunnel passing along a chord as shown in figure. Assume Earth's uniformity is not disturbed by drilling the tunnel.



[E = Total mechanical energy, R = Radius of earth,
g = Acceleration due to gravity at surface of earth]
Choose the correct options:

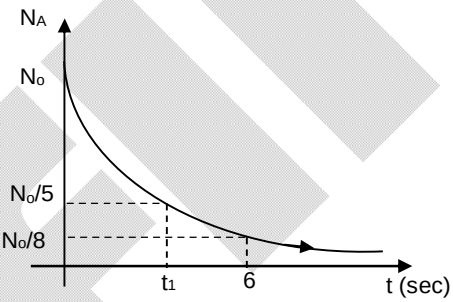
(A) $E_A = E_B$

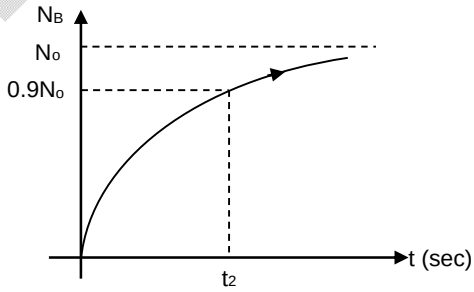
(B) $E_A > E_B$

(C) Speed of B when A reaches initial position of B is $\frac{3}{4}\sqrt{gR}$

(D) When B reach O_1, the relative speed of A w.r.t. B is less than escape speed from Earth's surface.

7. In a radioactive decay process A decays into B. Two graphs of number of nuclei of A and B verses time are shown. Mark the correct statements.





(A) $t_2 = t_1 = 4\,\text{sec}$

(B) $t_2 - t_1 = 2\,\text{sec}$

(C) $E_1 = 2\log_2 5\,\text{sec}$

(D) Half life of A is 2 sec.

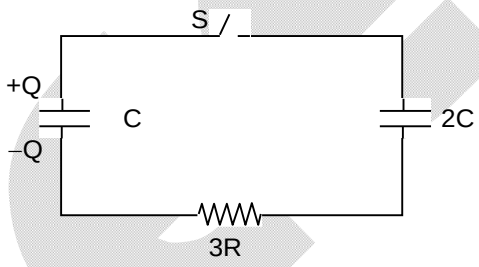
SECTION – B

(Numerical Answer Type)

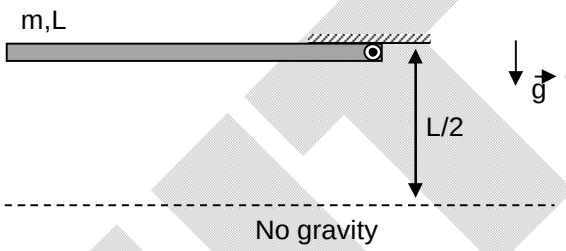
*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

8. A particle is executing SHM with time period T and amplitude A along a horizontal direction along x-axis. When it is at $+\frac{A}{2}$ from equilibrium position and moving towards extreme position (right) it receives an impulse which doubles its velocity without changing its direction. The time taken from this instant till it comes to rest for the first time is found to be $t=\frac{T}{2\pi}\cos^{-1}\left[\frac{1}{\sqrt{k}}\right]$sec. Find k.

9. In the circuit shown S is initially open and charge on capacitor C is Q as shown. At t = 0, switch is closed. The time after which charge on both capacitors are equal is found to be $t = KRC\ell n2$. Find the value of k.

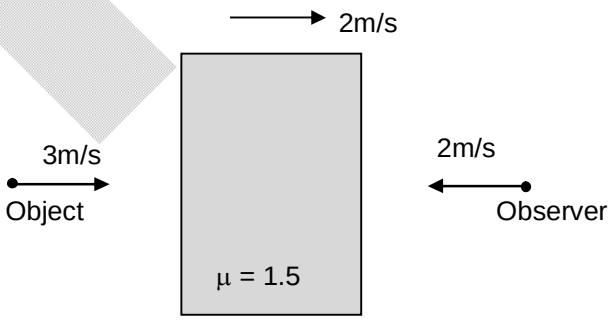


10. Consider a case where there is no. gravity below the broken line. Above the broken line gravity is unchanged, i.e; g vertically downward. If a hinged rod is released from horizontal position, then its angular speed when it becomes vertical is $\sqrt{\frac{ng}{4L}}$, find n.



11. A small ball of mass m carrying positive charge q is dropped under gravity in uniform horizontal magnetic field $\vec{B}$. The vertical depth of the deepest point of its path from initial position is $\frac{(k-1)m^2g}{q^2B^2}$. Find k.

12. A glass slab $(\mu = 1.5)$, a point object and an observer are moving in air as shown. Velocity of image w.r.t. observer is found to be n m/s. Find n.



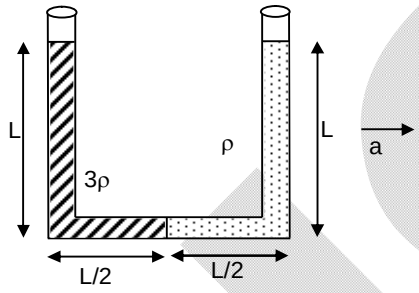
13. A mass of 1 kg executes oscillatory motion along one dimension under a restoring force which is proportional to displacement from mean position and a resisting force characterized by damping constant of 0.6 N sm^{-1}. If the value of force constant of restoring force in N/m is K so that motion will be critically damped, find 100K.

SECTION – C
(Numerical Answer Type)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Paragraph for Question Nos. 14 and 15

Figure shows a U-tube having two immiscible liquids of density 3ρ and ρ as shown in figure. The system is accelerated towards right with acceleration $\vec{a}$ such that level of liquid in both arms are equal in length.



[Take g = 10 m/s^2, $\rho = 10^2$ kg/m^3, L = 10 cm]

14. Value of a is _____ m/s^2.

15. Now the moving system is suddenly stopped. The gauge pressure at interface of both liquids just after that is _____N/m^2.

Paragraph for Question Nos. 16 and 17

A simple harmonic plane progressive wave is travelling along x-axis having amplitude A and $\omega = 10\pi \text{ rad/sec}$. At t = 2sec, particle at x = 2m (P) and x = 4m(Q) are moving downward and upward direction respectively at the position $y = -\frac{\sqrt{3}A}{2}$. Assume P and Q points are within one wavelength of the wave.

16. Wavelength of the wave is _____ m.

17. Time when maximum power transfer takes place at x = 3.5 m is found to be $\frac{A}{24}$ second. The value of A is _____.

Chemistry

PART – II

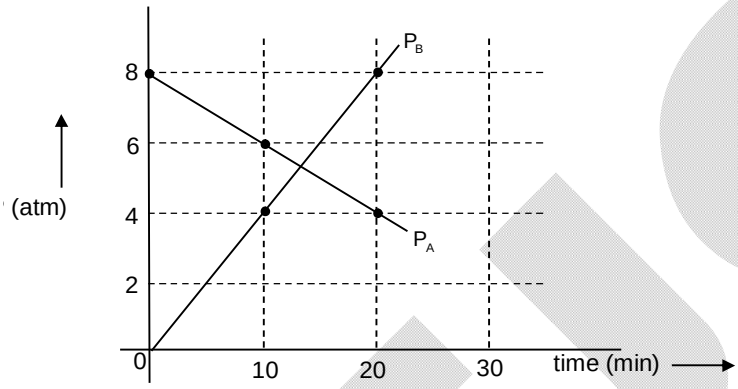
SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

18. Find the $E^0_{BrO_3^-/Br_2}$ in acidic medium from following data $BrO_3^- \xrightarrow[H^+]{E^0=0.54V} BrO^- \xrightarrow[H^+]{E^0=0.45V} \frac{1}{2}Br_2$

(A) – 0.059 V
(B) – 0.52 V
(C) – 0.828 V
(D) 0.0456 V

19. In reaction $A_{(g)} \longrightarrow B_{(g)}$ following observation is made.



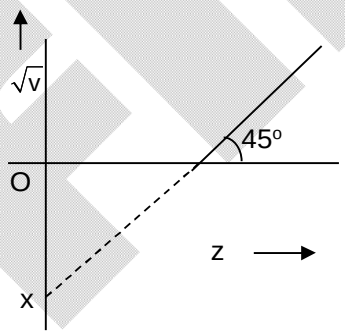
Then the incorrect statement is

(A) chemical reaction should be $\frac{1}{2}A_{(g)} \longrightarrow B_{(g)}$

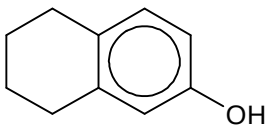
(B) $t_{\frac{1}{2}}$ of reaction is independent of initial pressure of A

(C) Reaction must be complex reaction
(D) Time for completion of reaction is 40 min.

20. In the graph between $\sqrt{v}$ and z for the Moseley's equation, the intercept OX is – 1 on $\sqrt{v}$ axis, then. What is frequency when z = 51.

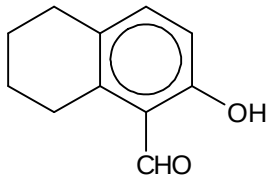


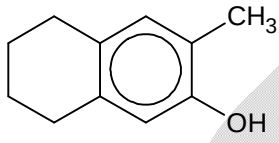
(A) 50 Hz
(B) 100 Hz
(C) 2500 Hz
(D) None of these

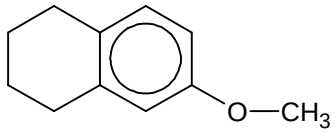
21. 

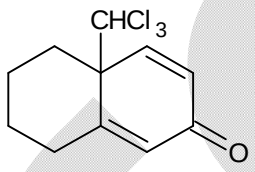
$\xrightarrow{CHCl_3 + KOH}$ A (aldehyde) + B, The product (not aldehyde)

B will be.

(A) 

(B) 

(C) 

(D) 

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

22. Which of the following statements is/are incorrect.

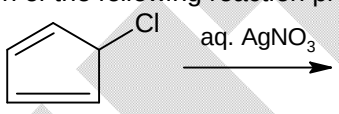
(A) In $\left[PtCl_2(NH_3)_4\right]^{2+}$, the cis form is optically active while trans form is optically inactive.

(B) In $\left[Fe(C_2O_4)\right]^{3-}$, geometrical isomers do not exist while optical isomers exist.

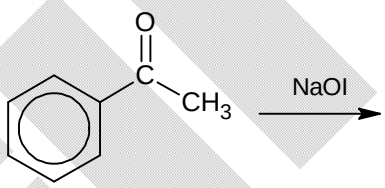
(C) $[Mabcd]^{n\pm}$ square planar complexes exhibit both optical and geometrical isomerism.

(D) In $[Mabcd]^{n\pm}$ tetrahedral complexes, optical isomerism is not observed

23. In which of the following reaction precipitate is formed.

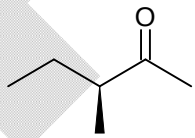
(A) 

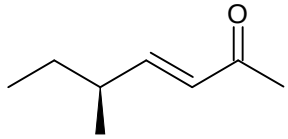
(B) $H_3C-C\equiv C-H \xrightarrow[\text{reagent}]{\text{Tollen's}}$

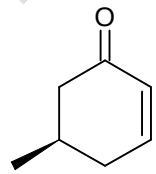
(C) 

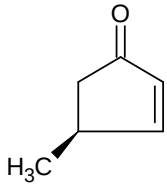
(D) $H_3C-\underset{\displaystyle Cl}{\underset{|}{CH}}-CH_3 \xrightarrow[\text{acetone}]{\text{NaI}}$

24. Each of the following compound is chiral. If a sample of pure enantiomer is treated with base, some of the compound rapidly converted into racemic mixture and same do not. Identify the compounds that rapidly converted into racemic mixture.

(A) 

(B) 

(C) 

(D) 

SECTION – B

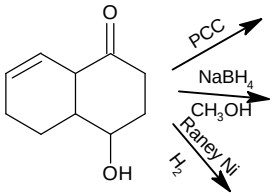
(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

25. K_a for HCN is 5×10^{-10} at 25°C. To maintain a constant pH of 9, the volume of 5M KCN solution to be added to 10 ml of 2 M HCN solution is
$\left(10^{-0.30}=0.5\right)\left(\log 5=0.7\right)$

26. How many of following will give black coloured precipitate with Na_2S solution?
$Pb(NO_3)_2$, $AgNO_3$, $NiCl_2$, $CdCl_2$, $CuCl_2$, $SnCl_4$, $ZnCl_2$, $CoCl_2$, $MnCl_2$, $SbCl_3$, $AlCl_3$

27. Total number of structures possible for C_4H_6.

28. 

sum of number of double bonds present in A, B and C will be.

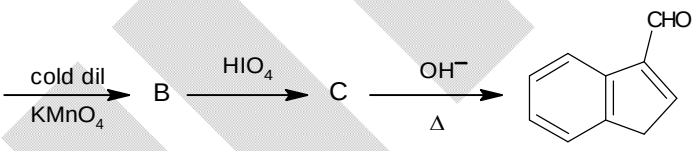
29. How many of following are optically active?

(i) Trans - $\left[Co(en)_2(NH_3)Cl\right]^{2+}$ (ii) Trans - $\left[Co(en)_2Br_2\right]$

(iii) $\left[Co(NH_3)Cl_3\right]$ (iv) Trans - $\left[Co(NH_3)_4Cl_2\right]^{+}$

(v) Trans - $\left[CoCl_2(C_2O_4)_2\right]^{3-}$ (vi) $\left[Co(en)_3\right]$

(vii) cis $-\left[Co(en)_2(NH_3)Cl\right]$ (viii) cis - $\left[Co(en)_2Br_2\right]$

30. 

'C' contain n number of sp^2 carbon and y number of sp^3 carbons. The value of |y – n| will be

SECTION – C

(Numerical Answer Type)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Paragraph for Question Nos. 31 and 32

Thallium exists in two different oxidation states Tl^+ and Tl^{3+}. Iodide ions can combine with iodine to form tri-iodide ions $\left(I_3^-\right)$ in aqueous solutions. The standard redox potentials for some relevant reactions are:

$Tl^+_{(aq)} + e^- \longrightarrow Tl_{(s)}$ $\quad E^0_1 = -0.336V$

$Tl^{3+}_{(aq)} + 3e^- \longrightarrow Tl_{(s)}$ $\quad E^0_2 = +0.728V$

$I_{2(s)} + 2e^- \longrightarrow 2I^-_{(aq)}$ $\quad E^0_3 = +0.540V$

The equilibrium constant for the reaction

$I_{2(s)} + I^-_{(aq)} \rightleftharpoons I^-_{3(aq)}$ $K_1 = 0.459$

Use $t = 25^oC$ throughout this problem.

31. The value of standard reduction potential in volt for the following half cell reaction is
$T\ell^{3+}_{(aq)} + 2e^- \longrightarrow T\ell^+_{(aq)}$

32. The cumulative complex formation constant for the reaction given below is $10^{35.7}$.
$T\ell^{3+}_{(aq)} + 4I^-_{(aq)} \rightleftharpoons T\ell I^-_{4(aq)}$
The value of natural logarithm of equilibrium constant ($\ln K_{eq}$) for the following reaction is approximately
$T\ell^+_{(aq)} + I^-_{3(aq)} + I^-_{(aq)} \rightleftharpoons T\ell I^-_{4(aq)}$

Paragraph for Question Nos. 33 and 34

The lead(II) oxide formed by roasting is reduced with more galena in a blast furnace to produce liquid lead and liberates sulfur dioxide gas.
A 1.045 g sample of lead rich galena is to be analyzed for purity and composition. Two-third of the sample is roasted in a limited air supply to give lead oxide. 66.2 mL of sulfur dioxide gas was liberated from the oxidation (when measured at 298 K and 1 atm)
The lead oxide was then reduced with the remaining galena sample in a blast furnace, and the liquid lead produced was cooled into an ingot weighing just 0.8663 g. (mol. wt. of galena = 239.2)

33. The purity of the galena (mass of lead sulfide per gram of galena) is

34. The produced lead oxide and remaining galena sample is reduced in blast furnace to get 0.8663 g lead. Then, what percent of the lead in the original galena sample was present as elemental lead?

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

35. Let z_1 and z_2 be non-zero complex numbers satisfying the equation, $z_1^2 - 2z_1z_2 + 2z_2^2 = 0$. The geometrical nature of the triangle whose vertices are the origin and the points representing z_1 and z_2 is:
(A) an isosceles right angled triangle
(B) a right angled triangle which is not isosceles
(C) an equilateral triangle
(D) an isosceles triangle which is not right angled

36. Let P_n denotes the number of ways of selecting 3 people out of n sitting in a two, if no two of them are consecutive and Q_n is the corresponding figure when they are in a circle. If $P_n - Q_n = 6$, then n is equal to:
(A) 8 (B) 9
(C) 10 (D) 12

37. A function y = f(x) satisfies $f''(x) = -\frac{1}{x^2} - \pi^2 \sin(\pi x); f(2) = \pi + \frac{1}{2}$ f(1) = 0. The value of $f\left(\frac{1}{2}\right)$ is:
(A) ln 2 (B) 1
(C) $\frac{\pi}{2} - \ln 2$ (D) $1 - \ln 2$

38. Through the focus of the parabola $y^2 = 2px (p > 0)$ a lines is drawn which intersects the curve at $A(x_1, y_1)$ and $B(x_2, y_2)$. The ratio $\frac{y_1x_1}{x_1x_2}$ equals:
(A) 2 (B) – 1
(C) – 4 (D) some function of P

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

39. Two whole numbers are randomly selected and multiplied. Consider two events E_1 and E_2 defined as:
E_1 : Their product is divisible by 5
E_2 : Unit's place in their product is 5.
Which of the following statement(s) is/are correct?
(A) E_1 is twice as likely to occur as E_2 (B) E_1 and E_2 are equiprobable.
(C) $P(E_2 / E_1) = \frac{1}{4}$ (D) $P(E_1 / E_2) = 1$

40. The value of θ lying between $-\frac{\pi}{4}$ and $\frac{\pi}{2}$ and $0 \le A \le \frac{\pi}{2}$ and satisfies the equation

$$\begin{vmatrix} 1+\sin^2 A & \cos^2 A & 2\sin 4\theta \\ \sin^2 A & 1+\cos^2 A & 2\sin 4\theta \\ \sin^2 A & \cos^2 A & 1+2\sin 4\theta \end{vmatrix} = 0$$, are:

(A) $A = \frac{\pi}{4}, \theta = -\frac{\pi}{8}$ (B) $A = \frac{3\pi}{8} = \theta$

(C) $A = \frac{\pi}{5}, \theta = -\frac{\pi}{8}$ (D) $A = \frac{\pi}{6}, \theta = \frac{3\pi}{8}$

41. Let f(x) = $\min\left(x^3, x^2\right)$ and $g(x) = [x]^2 + \sqrt{\{x\}^2}$, where [x] denotes the greatest integer and {x} denotes the fractional part function. Then which of the following holds?

(A) f is continuous for all x. (B) g is discontinuous fall all $x \in I$

(C) f is differentiable for all $x \in (1, \infty)$ (D) g is not differentiable for all $x \in I$

SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

42. Let P_1 be a plane passing through two points A(1, 2, 0) and B(2, –1, 1) and perpendicular to $\vec{r} \cdot (i + j - 3k) = 1$. Let L be a line through (1, 1, 1) and perpendicular to line of intersection of planes P_1 and P_2 where equation of plane P_2 is $\vec{r} \cdot (i + j + k) = 5$. If the equation of line L is $\frac{x+1}{-1} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$, then find the value of $\left(\frac{2y_1 + z_1}{bc}\right)$.

43. Let $I = \int_0^1 \left(1-x^{50}\right)^{99} x^{100} dx$ and $J = \int_0^1 \left(1-x^{50}\right)^{100} x^{100} dx$. If $\frac{I}{J}$ is equal to $\frac{m}{n}$, where m and n are co-prime, then find the value of $\left(\frac{m-n-1}{20}\right)$.

44. If $f(x) = \ln\left(1+x^2\right) + \tan^{-1} x, x > 0$ and $g(x) = f^{-1}(x)$, then find the value of $\left|27g''\left(\ln 2e^{\frac{\pi}{4}}\right)\right|$ where g"(x) denotes second derivative of g(x).

45. If $\sum_{r=1}^{10} \tan^{-1}\left(\frac{3}{9r^2+3r-1}\right) = \cot^{-1}\left(\frac{m}{n}\right)$ (where m and n are coprime), then find (2m + n).

46. Let $f(x) = 1 - x + x^2 - x^3 + ... + x^{16} - x^{17} = a_0 + a_1(1+x) + a_2(1+x)^2 + ... + a_{17}(1+x)^{17}$. Find the value of a_2.

47. The roots α and β of a quadratic equation are the square of two consecutive natural numbers. The geometric mean of the two roots is 1 greater than the difference of the roots. If α and β lies between the roots of $x^2 - cx + 1 = 0$, then find the minimum integral value of c.

SECTION – C
(Numerical Answer Type)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Paragraph for Question Nos. 48 and 49

Let f(x) be a polynomial function of degree 2satisfying.

$$\int \frac{f(x)}{x^3-1}dx = \ln\left|\frac{x^2+x+1}{x-1}\right| + \frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + C$$

where C is indefinite integration constant.

48. The value of f(5) is equal to:

49. Let $\int \frac{1-6\operatorname{cosec} x}{6+f(\sin x)}d(\sin x) = g(x)+K,$ where g(x) contains no constant term. Then $\lim_{t\to\pi/2} g(t)$ is equal to (where K is indefinite integration constant):

Paragraph for Question Nos. 50 and 51

Let $f(t) = \sum_{k=0}^{\infty} \frac{1}{(t+2)^k}, t > -1$

50. Value of $\sum_{r=1}^{\infty} \frac{r}{(f(0))^{r-1}}$ is:

51. Value of $\sum_{k=1}^{\infty} \frac{f(x)}{k(k+2)}$ is: